Simulation Lifecycle Monitoring Module (SLMM)

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Independent Methodological Authority

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Canonical Definition

Simulation Lifecycle Monitoring Module (SLMM) is the first Core Module of the Simulation Lifecycle & Resimulation Standard (SLRS) within SIMULOS® — Simulation Governance Architecture.

SLMM governs the continuing observation of conditions capable of materially affecting the relevance, applicability, representativeness, evidential meaning, or continued use of an existing simulation and its governed results.

The Module establishes a structured relationship between the simulation basis that existed when simulation evidence was produced and the conditions that exist as the simulation continues through its lifecycle.

SLMM does not determine whether observed change constitutes Simulation Drift, nor whether resimulation is required.

Its task is foundational:

to ensure that material changes capable of affecting simulation relevance can become visible rather than allowing a simulation to remain operationally influential after its governing basis has silently changed.

Core Question

Which conditions must remain monitored so that material changes affecting simulation relevance can be detected throughout the simulation lifecycle?

Architectural Position

SLMM is the first module within SLRS.

It follows the completed active simulation lifecycle:

Simulation Object → Purpose & Scope → Reality Representation → Assumptions → Model & Environment → Inputs & Data → Scenario Design → Execution → Output Governance → Simulation-to-Reality Correlation → Simulation Evidence & Reliance Boundary

and begins lifecycle governance.

SLMM precedes:

SDRM — Simulation Drift Recognition Module.

The relationship is:

Lifecycle Monitoring → Drift Recognition.

SLMM observes and structures relevant change.

SDRM determines whether that change creates a material divergence between the existing simulation basis and the conditions to which that simulation remains applied.

Internal Flow Position

Governed Simulation Lifecycle Basis → Monitoring Context Definition → Change-Sensitive Condition Identification → Lifecycle Observation → Change Signal Detection → Material Change Candidate Registration → Simulation Relevance Review Input → SDRM — Simulation Drift Recognition Module Operational Space

SLMM governs the space between:

an established governed simulation basis

and

subsequent changes capable of affecting that basis.

This includes monitoring relevant change affecting: Simulation Object identity; object version; object configuration; object state; simulation purpose; simulation scope; represented reality; abstraction; fidelity requirements; assumptions; model applicability; model version; simulation environment; parameterization; inputs; data distributions; temporal data relevance; input uncertainty; scenario relevance; scenario coverage; observed operational conditions; simulation-to-reality correlation; evidence applicability; evidence weight; reliance conditions; known limitations; previously unknown conditions.

Operational Scope

SLMM applies wherever simulation results, evidence, models, scenarios, or simulation-derived conclusions continue to be retained, reused, referenced, compared, relied upon, or maintained after their initial production.

It may govern lifecycle monitoring for: individual simulations; simulation models; repeated simulation programs; scenario portfolios; Monte Carlo studies; digital twins; engineering simulations; scientific simulations; safety simulations; infrastructure simulations; mission simulations; economic simulations; AI simulations; autonomous-system simulations; large-scale simulation campaigns; long-duration simulation programs; future simulation technologies.

The Module remains technology-neutral.

Its concern is not how frequently a simulator executes.

Its concern is whether the conditions supporting simulation relevance remain sufficiently observable over time. Simulation Completion ≠ Lifecycle Completion

SLMM establishes the distinction:

Simulation Execution Completion ≠ Simulation Lifecycle Completion

A simulation run may end.

A simulation campaign may end.

The simulator may be switched off.

But the simulation results may continue influencing: engineering; design; planning; scientific reasoning; safety assessment; mission preparation; system development; validation; operational decisions; deployment; future simulation campaigns.

Lifecycle governance therefore continues for as long as simulation-derived information remains materially relevant or relied upon.

Lifecycle Monitoring Basis

SLMM requires a Simulation Lifecycle Monitoring Basis to identify which parts of the governed simulation lifecycle are capable of materially affecting continuing relevance.

The monitoring basis may reference:

Simulation Object → what was simulated.

Purpose & Scope → why and within what boundaries it was simulated.

Reality Representation → which reality was represented and how.

Assumptions → what conditions were presumed.

Model & Environment → which model and simulation environment generated the behavior.

Inputs & Data → which information basis entered the simulation.

Scenarios → which conditions were tested.

Execution → how the simulation was actually executed.

Outputs → what was produced.

Simulation-to-Reality Correlation → what relationship with reality was established.

Evidence & Reliance Boundary → what the simulation-derived evidence was permitted to support.

Lifecycle monitoring does not require every element to be monitored equally.

It requires monitoring to remain proportionate to the conditions capable of materially changing simulation relevance.

Change-Sensitive Condition Principle

Not every characteristic of a simulation requires continuous monitoring.

SLMM identifies Change-Sensitive Conditions:

conditions whose material change could alter the relevance, applicability, representativeness, evidential meaning, or legitimate reliance of an existing simulation.

Examples may include: object redesign; software update; hardware replacement; environmental change; changed operational profile; new observed behavior; changed data distribution; new failure mode; revised assumption; model discovery; changed mission condition; new regulatory operating constraint; new physical observation; changed scenario prevalence; newly discovered edge case; degraded simulation-to-reality correlation.

The monitoring burden should therefore follow material relevance, not indiscriminate observation.

Reality Change Monitoring

A simulation may remain unchanged while reality changes around it.

SLMM therefore requires relevant changes in represented or relied-upon reality to remain observable where those changes can affect simulation relevance.

Examples may include: environmental change; infrastructure change; human behavioral change; operational practice change; traffic pattern change; climate condition change; market structure change; mission environment update; newly observed physical behavior; new operational interactions.

ORA™ continues to govern Operational Reality itself.

SLMM governs whether changes in relevant reality are being observed for their potential effect on simulation relevance.

Simulation Object Change Monitoring

A simulation may become stale because the Simulation Object itself changes.

SLMM may monitor: new object versions; configuration changes; component replacement; software changes; hardware changes; changed interfaces; changed capabilities; changed constraints; changed operational modes; changed dependencies.

A simulation of Object Version V3 does not automatically remain representative of Object Version V7.

Object evolution must therefore remain connected to lifecycle monitoring.

Assumption Change Monitoring

Assumptions that were reasonable when a simulation was constructed may later become questionable, obsolete, contradicted, or unnecessary.

SLMM may therefore monitor: newly available evidence affecting assumptions; observed contradiction; changed uncertainty; revised scientific understanding; changed operational conditions; dependence on assumptions no longer applicable.

SLMM does not govern the assumptions themselves.

That remains under SAS — Simulation Assumption Standard.

It monitors changes capable of affecting their continued applicability.

Model and Environment Change Monitoring

Simulation models and environments may evolve independently of the simulation evidence already produced.

SLMM may monitor: model replacement; model revision; environment updates; parameterization changes; discovered model defects; corrected physical relationships; changed solver behavior; updated simulation infrastructure; changed environmental representations.

A newer model does not automatically invalidate earlier simulations.

But material differences must become visible for lifecycle assessment.

Input and Data Change Monitoring

Simulation relevance may degrade when the distributions or conditions represented by simulation inputs materially change.

SLMM may monitor: new operational data; changed distributions; changed populations; changed environmental data; temporal shifts; new sensor characteristics; newly available empirical data; changed uncertainty; changed input-generation methods.

SIDS continues to govern simulation inputs and data at the simulation-input boundary.

SLMM monitors subsequent changes that may affect the relevance of previously generated simulation results.

Scenario Relevance Monitoring

Scenario sets may become incomplete over time.

SLMM may identify changes such as: new operational scenarios; newly observed edge cases; changed scenario frequency; emerging stress conditions; newly relevant interactions; previously unknown failure combinations; changed environmental combinations.

The discovery of a new scenario does not automatically invalidate existing simulations.

It may, however, reveal that existing scenario coverage no longer represents the relevant problem space sufficiently.

Correlation Change Monitoring

Simulation-to-reality correlation is not necessarily permanent.

New real-world observations may show: stronger correlation; weaker correlation; systematic deviation; condition-specific divergence; temporal degradation; newly discovered mismatch.

SLMM therefore monitors information capable of changing the existing correlation basis established through SRCS.

It does not recalculate correlation.

That remains within Simulation-to-Reality Correlation Standard (SRCS).

Evidence Relevance Monitoring

Simulation-derived evidence may remain technically unchanged while its relevance changes.

SLMM may monitor conditions capable of affecting: evidence qualification; evidential weight; applicability; reliance boundaries; use limitations; temporal relevance.

Where a material lifecycle change affects simulation evidence, the relevant SERBS governance may require reassessment.

SLMM does not independently redefine evidential weight or reliance.

Change Signal

SLMM introduces the Simulation Lifecycle Change Signal.

A Change Signal is:

an observed or reported change that may materially affect the continuing relevance of a governed simulation basis.

A Change Signal is not yet Simulation Drift.

It is an input for further assessment.

Possible sources include: operational observations; engineering change records; new measurements; updated datasets; new scientific findings; system version changes; mission updates; incident findings; model corrections; new scenario discoveries; changed environmental conditions; changed correlation results.

Material Change Candidate

Where a Change Signal has a plausible relationship to simulation relevance, SLMM may register it as a Material Change Candidate.

The distinction is:

Change Signal → something changed.

Material Change Candidate → the change may matter to the simulation.

Simulation Drift → the change has produced a material divergence affecting the governed simulation basis.

The final determination of Simulation Drift belongs to SDRM.

Monitoring by Dependency

Simulation relevance may be affected indirectly.

For example:

Component Change → changes system behavior → changes object configuration → affects scenario behavior → changes simulation applicability.

SLMM therefore permits lifecycle monitoring across material dependency chains rather than monitoring only direct changes to the simulator.

This is especially important for complex systems where a seemingly small change may propagate into a simulation-relevant difference.

Event-Driven Monitoring

Lifecycle monitoring may be event-driven.

Relevant events may include: object update; software release; hardware change; new operational dataset; incident; newly observed failure; mission redesign; environmental change; assumption challenge; model correction; correlation deterioration; regulatory operating change.

An event may initiate a simulation relevance review without waiting for a scheduled lifecycle review.

Periodic Monitoring

Where simulation relevance may degrade gradually or where material change is not tied to a discrete event, organizations may use periodic monitoring.

The appropriate interval may depend upon: rate of object change; rate of environmental change; consequence of stale simulation reliance; simulation purpose; operational volatility; evidence importance; availability of new reality information.

SLMM does not impose one universal review interval.

Continuous Monitoring

Certain simulation contexts may justify continuous or near-continuous lifecycle monitoring.

Examples include: digital twins; rapidly changing autonomous systems; high-frequency operational environments; continuously updated models; adaptive systems; dynamic infrastructure environments.

Continuous monitoring does not imply continuous resimulation.

It provides a higher-frequency basis for detecting potentially material change.

Monitoring Proportionality Principle

SLMM rejects the assumption that all simulations require identical monitoring intensity.

Monitoring should be proportionate to: rate of relevant change; consequence of outdated reliance; simulation reuse; operational importance; uncertainty; dependence on volatile assumptions; dependence on changing reality; expected lifecycle duration.

A one-time exploratory simulation may require minimal lifecycle monitoring.

A simulation supporting years of safety-critical operational reliance may require substantially stronger monitoring.

Dormant Simulation Principle

A simulation may remain unused for a period and later return to active use.

Dormancy must not create an assumption of continuing relevance.

Before material reuse, the simulation may require review against accumulated lifecycle changes.

Therefore:

Stored Simulation ≠ Current Simulation Basis.

Monitoring at Scale

Large organizations may maintain thousands or millions of simulation artifacts and evidence objects.

SLMM must therefore support lifecycle monitoring at scale.

Monitoring may operate through: simulation families; object families; model versions; scenario families; evidence classes; dependency relationships; change-event propagation; lifecycle status registers.

A material change affecting one shared model may therefore identify many dependent simulation results requiring review.

Lifecycle Monitoring State

SLMM may assign a lifecycle monitoring state such as: 1. Active Monitoring

Relevant lifecycle conditions are currently monitored. 2. Reduced Monitoring

Monitoring continues at a reduced level because active reliance or change exposure is limited. 3. Dormant

The simulation is retained but not currently subject to active reliance. 4. Review Required

A material Change Signal or accumulated lifecycle change requires assessment. 5. Monitoring Suspended

Required monitoring cannot currently be performed sufficiently. 6. Lifecycle Exit Candidate

The simulation may no longer justify continued monitoring and may require retirement assessment.

These are governance states rather than universal technical classifications.

Lifecycle Traceability

SLMM requires material lifecycle changes to remain traceable to the simulation assets potentially affected by them.

A lifecycle trace may connect:

Change Event → Affected Reality Condition → Affected Simulation Basis Element → Affected Simulation → Affected Outputs → Affected Evidence → Potential Reliance Impact

This relationship enables downstream drift recognition and proportionate resimulation decisions.

SLMM Boundary with SDRM

SLMM identifies and structures potentially relevant lifecycle change.

SDRM determines whether the change constitutes material Simulation Drift.

Therefore:

Monitoring Signal ≠ Drift Determination.

SLMM Boundary with SRGM

SLMM does not determine whether resimulation must occur.

That decision belongs to:

SRGM — Resimulation Trigger Module.

The lifecycle sequence is:

Monitoring → Drift Recognition → Trigger Determination.

SLMM Boundary with ORA™

ORA™ governs actual Operational Reality.

SLMM may consume relevant reality-change information to determine whether simulation relevance requires review.

It does not redefine, reconstruct, or govern Operational Reality itself.

SLMM Boundary with SERBS

SERBS governs simulation-derived evidence, evidential weight, reliance boundaries, and use limitations.

SLMM monitors changes capable of affecting those governed evidence conditions.

It does not independently redefine evidence qualification, weight, or permitted reliance.

SLMM Boundary with VALIDOS®

VALIDOS® governs validation and revalidation.

SLMM governs simulation lifecycle monitoring.

A lifecycle change detected by SLMM may eventually contribute to revalidation processes, but:

Simulation Lifecycle Review ≠ Validation Reassessment.

SLMM Boundary with INTEGROS®

SLMM does not govern general lifecycle integrity.

INTEGROS® continues to govern integrity properties of objects, states, evidence, and processes.

SLMM governs only lifecycle monitoring specifically required to preserve the continuing relevance of simulation.

SLMM Boundary with ADIT®

ADIT® may audit whether lifecycle records, change evidence, and monitoring activities are observable, preserved, and verifiable.

SLMM owns the simulation-specific monitoring logic.

Lifecycle Monitoring ≠ Lifecycle Audit.

Governance Outputs

SLMM may produce: Simulation Lifecycle Monitoring Basis Change-Sensitive Condition Register Simulation Lifecycle Monitoring Profile Simulation Lifecycle Change Signal Material Change Candidate Record Simulation Object Change Link Reality Change Link Assumption Change Link Model & Environment Change Link Input & Data Change Link Scenario Relevance Change Link Correlation Change Link Evidence Relevance Change Link Simulation Dependency Change Record Lifecycle Monitoring State Simulation Relevance Review Input Simulation Lifecycle Monitoring Record

These outputs provide the governed input basis for:

SDRM — Simulation Drift Recognition Module.

Enterprise Application

SLMM enables organizations to answer a lifecycle question that conventional simulation programs frequently leave implicit:

How will we know when the simulation we trusted yesterday no longer represents the problem we face today?

An organization may use SLMM to detect when: a simulated product has changed configuration; new operational behavior has emerged; a mission environment has been revised; new physical measurements challenge earlier assumptions; an autonomous system receives a significant software update; operational data distributions change; new edge cases appear; a model defect is discovered; simulation-to-reality correlation deteriorates; simulation evidence remains in organizational use long after its original basis has changed.

SLMM does not force the organization to rerun every simulation whenever something changes.

Instead, it creates the governed observation layer necessary to determine whether the change matters at all.

This allows later modules to distinguish between:

irrelevant change, material drift, targeted resimulation, broad resimulation, replacement, and retirement.

Foundational Principles

Simulation completion is not simulation lifecycle completion.

A simulation may remain computationally unchanged while becoming operationally stale.

Lifecycle monitoring must follow the conditions that give simulation results their meaning.

Not every change is material to simulation relevance.

A detected change is not automatically Simulation Drift.

Monitoring intensity should be proportional to the consequence and likelihood of simulation obsolescence.

Stored simulation results do not automatically remain current simulation evidence.

Reality change, object change, assumption change, model change, data change, scenario change, and correlation change may all affect continuing simulation relevance.

Continuous monitoring does not require continuous resimulation.

A simulation cannot remain governably relevant over time if material changes to the basis supporting its relevance are allowed to remain invisible.

Canonical Closing Statement

Simulation Lifecycle Monitoring Module establishes the lifecycle observation layer of the Simulation Lifecycle & Resimulation Standard by maintaining structured visibility over changes capable of affecting the continuing relevance, applicability, representativeness, evidential meaning, and legitimate use of governed simulations. Through governance of monitoring bases, change-sensitive conditions, object change, reality change, assumption change, model and environment change, input and data change, scenario relevance, correlation change, evidence relevance, dependency propagation, change signals, material change candidates, monitoring states, and lifecycle traceability, SLMM prevents simulation results from remaining operationally influential merely because the conditions that once supported them are no longer being examined. By separating lifecycle observation from Simulation Drift recognition and resimulation decisions, SLMM creates the first governed control point of the recursive SIMULOS® lifecycle and provides the structured change basis required by SDRM. A simulation can remain governably relevant over time only when material changes capable of altering its meaning can first become visible.

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